N 805. 1. $9x^2+16y^2-54x+64y+1=0,$

$9(x-3)^2+16(y+2)^2-81+1-64=0,$

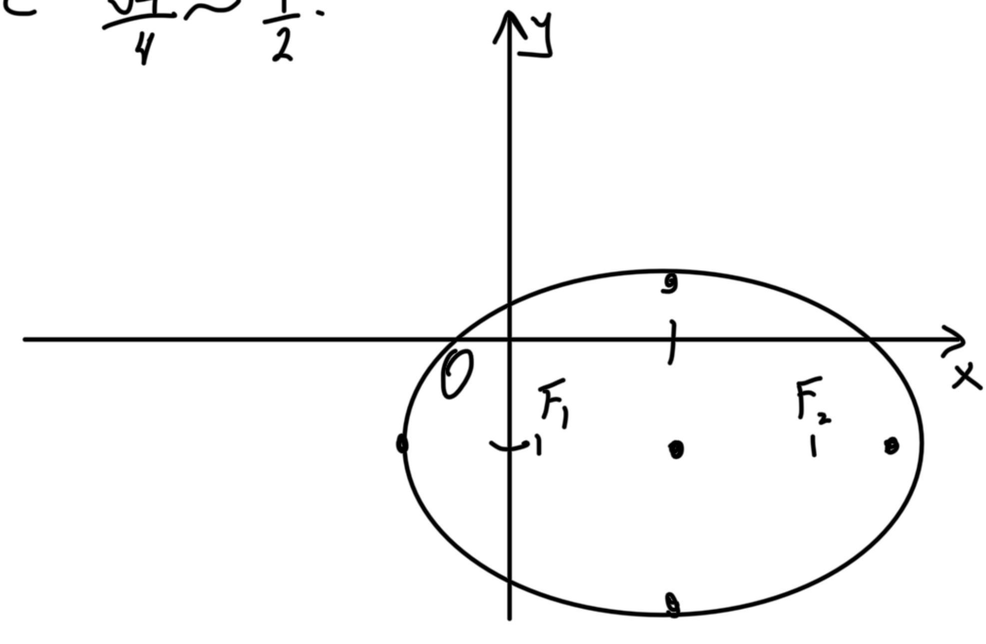
$9(x-3)^2+16(y+2)^2=144,$

$\begin{cases} x'=x-3, \\ y'=y+2 \end{cases} \Rightarrow \quad 9x'^2+16y'^2=144,$

$\frac{x'^2}{4^2}+\frac{y'^2}{3^2}=1,$ ← эллипсоид

$c=\sqrt{16-9}=\sqrt{7};$

$\varepsilon=\frac{\sqrt{7}}{4}\approx\frac{1}{2}.$




2. $4x^2-y^2-16x-6y+3=0,$

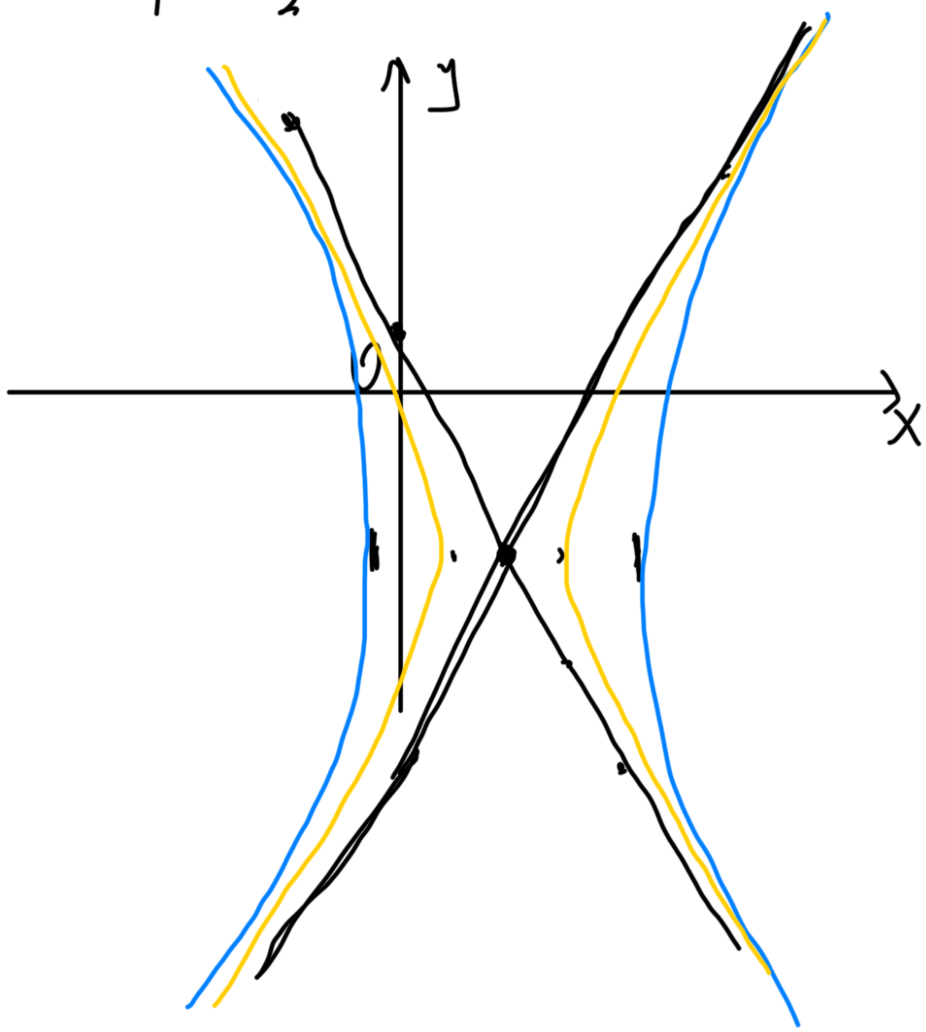
$4(x-2)^2-(y+3)^2+3-16+9=0,$

$\begin{cases} x'=x-2, \\ y'=y+3 \end{cases} \quad 4x'^2-y'^2=4,$

$\frac{x^2}{1^2} - \frac{y^2}{2^2} = 1,$ ← гипербола

$C = \sqrt{1+4} = \sqrt{5}$

$y = \pm 2x$




3) $3y^2 - 12x - 6y + 11 = 0,$ $\quad \delta = \begin{vmatrix} 0 & 0 \\ 0 & 3 \end{vmatrix} = 0$

⇓

парабола

$3(y-1)^2 - 3 + 11 - 12x = 0,$

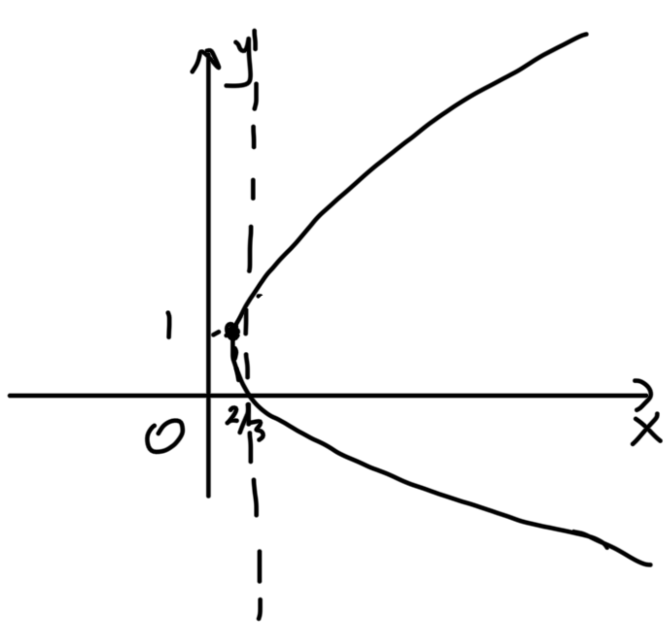
$\begin{cases} y' = y - 1 \\ x' = x - \frac{2}{3} \end{cases} \Rightarrow 3y'^2 = 12x - 8$

$y'^2 = 4x - \frac{8}{3}.$

$y'^2 = 4\left(x - \frac{2}{3}\right);\quad x' = x - \frac{2}{3}$

$y'^2 = 2 \cdot 2x';\quad p = 2$

$F(1;0)$
$d = x+1$



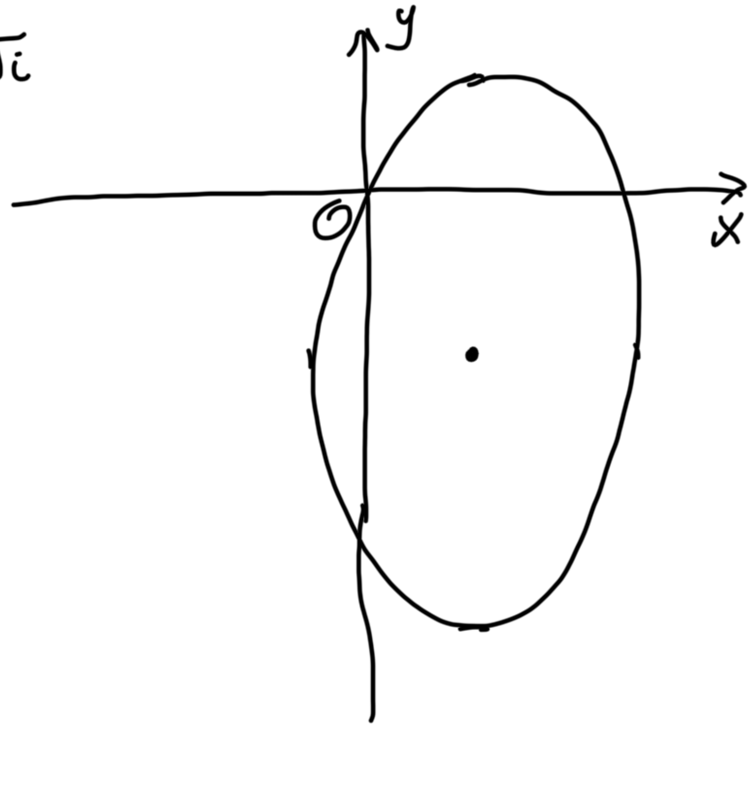

4) $25x^2+9y^2-100x+54y-44=0,$

$25(x-2)^2+9(y+3)^2-100-81-44=0,$

$\begin{cases} x'=x-2 \\ y'=y+3 \end{cases} \Rightarrow 25x'^2+9y'^2=225,$

$\frac{x'^2}{3^2}+\frac{y'^2}{5^2}=1, \leftarrow$ эллипс

$c=\sqrt{-3}=3\sqrt{i}$



б) $3x^2+12x+16y-12=0$ $\qquad J=\begin{vmatrix}3&0\\0&0\end{vmatrix}=0$

$3(x+2)^2=\cancel{12}-16y,$

$(x+2)^2=-\frac{16}{3}y+4,$

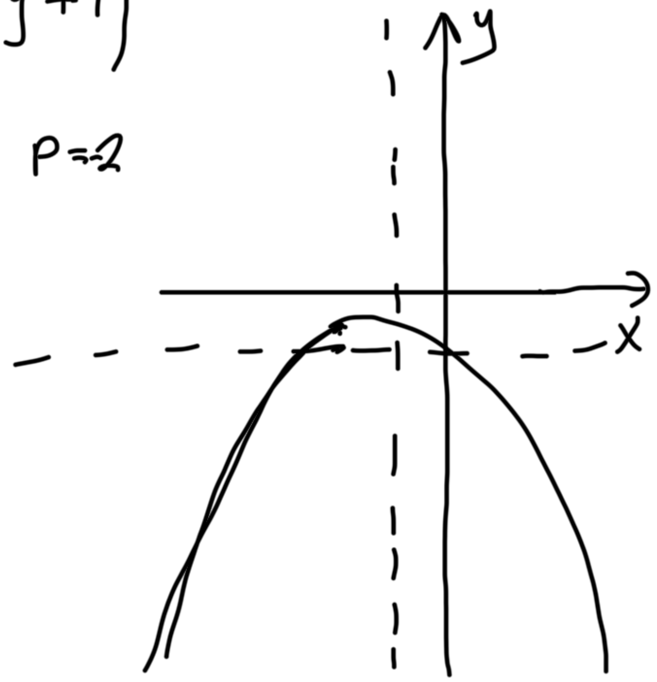
$(x+2)^2=-4\left(\frac{4}{3}y+1\right)$

$$\begin{cases}x'=x+2\\ y'=\frac{4}{3}y+1.\end{cases}$$

$x'^2=-4y'.$ $\quad p=-2$

$F(-1;0)$

$d=x-1.$




$$\begin{array}{ccccccccccc} & a_{11} & a_{22} & a_{33} & a_{12} & a_{23} & a_{13} & a_{14} & a_{24} & a_{34} & a_{44}\end{array}$$

№ 1046 1) $4x^2+y^2+4z^2-4xy+4yz-8zx-28x+2y+16z+45=0$

$J_1=9;$ $\quad J_2=\begin{vmatrix}4&-2\\-2&1\end{vmatrix}+\begin{vmatrix}4&-4\\-4&4\end{vmatrix}+\begin{vmatrix}1&2\\2&4\end{vmatrix}=0+0+0=0$

$J_3=\begin{vmatrix}4&-2&-4\\-2&1&2\\-4&2&4\end{vmatrix}=\cancel{16}-\cancel{16}+\cancel{16}-\cancel{16}-16-16=-32$

$$\begin{vmatrix}a_{11}&a_{12}&a_{13}&a_{14}\\a_{21}&a_{22}&a_{23}&a_{24}\\a_{31}&a_{32}&a_{33}&a_{34}\end{vmatrix}\longleftrightarrow\begin{vmatrix}x^2&2xy&2xz&x\\&y^2&2yz&y\\&&z^2&z\\&&&C\end{vmatrix}$$

$$\begin{vmatrix} a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

$$\begin{vmatrix} x^2 & 2xy & 2xz & 2x \\ & y^2 & 2yz & 2y \\ & & z^2 & 2z \\ 0 & & & c \end{vmatrix};\quad J(\lambda)=$$

$$-\lambda^3+9\lambda^2-32=0$$

$$J(\lambda) = -\lambda^3 + \lambda^2 J_1 - \lambda J_2 + J_3.$$

$$J(\lambda) = \begin{vmatrix} a-\lambda & & \\ & \beta-\lambda & \\ & & \gamma-\lambda \end{vmatrix} = (a-\lambda)(\beta-\lambda)(\gamma-\lambda).$$

$$J_1 = a_{11} + a_{22} + a_{33}$$

$$J_2 = \begin{vmatrix} a_{11} & a_{12} \\ & a_{22} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ & a_{33} \end{vmatrix}$$

$$J_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{vmatrix}.$$

$$J_4 = \begin{vmatrix} 4 & -2 & -4 & -14 \\ & 1 & 2 & 1 \\ & & 4 & 8 \\ & & & 45 \end{vmatrix} =$$

T(см.)

$$J_4 = \begin{vmatrix} 1 & 1 & 2 & 3 & 4 \\ 2 & & & & \\ 3 & & & & \\ 4 & & & & \end{vmatrix} = \begin{vmatrix} 4 & -2 & -4 & -14 \\ -2 & 1 & 2 & 1 \\ -4 & 2 & 4 & 8 \\ -14 & 1 & 8 & 45 \end{vmatrix} =$$

$$\lambda^3 - 9\lambda^2 + 32 = 0, \qquad \lambda^3 = 9\lambda^2$$

$-2: \; -8-30+32 \qquad \lambda = 9\lambda$

$$\boxed{\lambda = 9}$$

$$J(\lambda) = (\alpha-9)(\beta-9)(\gamma-9) = (\alpha\beta+81-9\alpha-9\beta)(\gamma-9) =$$

$$= \alpha\beta\gamma$$